# 1 The Research Process for Rogue Waves

## 1.1 The Research Process for Rogue Wave Phenomenon

Rogue wave is also named as freak wave, monster wave, extreme wave, killer wave, giant wave, etc. It is difficult to give a full explanation of rogue wave, due to its complex phenomenon. Figure 1.1 shows us the dramatic appearance of the rogue wave which was downloaded from the Internet. The authority statement in oceanography is that it develops from the ocean suddenly with high amplitude, which appears from nowhere and disappears without a trace. Nowadays, such nonlinear phenomenon has been observed in nonlinear optics, Bose–Einstein condensates (BEC), atmospherics, superfluid and even finance. It has aroused the attention of scientists from oceanography, physics and other nonlinear fields, since the term "freak rogue waves" was first coined by Draper [1] in 1965.

For centuries, rogue wave has been seen as the sea legend and a part of the marine folk culture. In the ocean, rogue waves, like a deep-sea monster, engulf sailors and ships without any trace, which caused a lot of maritime disasters.

In 1933, in the Pacific Northwest, USA warship *Ramapo* encountered the tallest rogue wave ever recorded, with 34 m (112 ft) height, which was visually measured by the crews on the deck [2].

Since 1952, in the Indian Ocean, at least 12 cases of ships encountering rogue wave have been recorded, near the Aga DeGeneres stream and along the coast of South Africa. One miserable case is that, on June 13, 1968, the tanker *World Glory* encountered a rogue wave, which broke the tanker into two parts (as seen in Figure 1.2) and caused the death of 22 crews, when the tanker was along the South African coast [3].

In 1966, the Michelangelo cruise ship, during the voyage from Italy to the United States, suddenly encountered a huge wave with 24 m height. The wave tore a hole in the superstructure, smashed the heavy glass, and made a crew member and two passengers die.

In 1978, *Munich*, a German barge, sank in the Atlantic Ocean. The twisted wreckage indicates that it was destroyed by a huge wave.

In 1980, a rogue wave with 25 m height was photographed by Philippe Lijour, the first mate of the French tanker *Esso Languedoc*, near the sea of the eastern port city Durban of South Africa (Azania). As seen in Figure 1.3, a huge water hole was observed clearly.

In 1984, a rogue wave attacked the 2/4-A oil platform of Norwegian Ekofisk oil field, which was located above average sea level of 20 m. In this accident, the wall of the platform's control room was destroyed, producing a standstill for 24 hours [4].

In 1986, a US warship *SS Spray* suffered three consecutive bursts of waves with approximately 25 m height in the sea of Charleston [5]. A highly asymmetrical rogue wave was also observed as shown in Figure 1.4, which was shot from the warship.

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**Figure 1.1:** A photograph of a rogue wave (source:http://de.weather-forecast.com/photos/41).

**Figure 1.2:** The tanker *World Glory* was destroyed by a rogue wave.

**Figure 1.3:** The rogue wave photographed from the tanker *Esso Languedoc*.

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**Figure 1.4:** The observation of a rogue wave from the warship *SS Spray*.

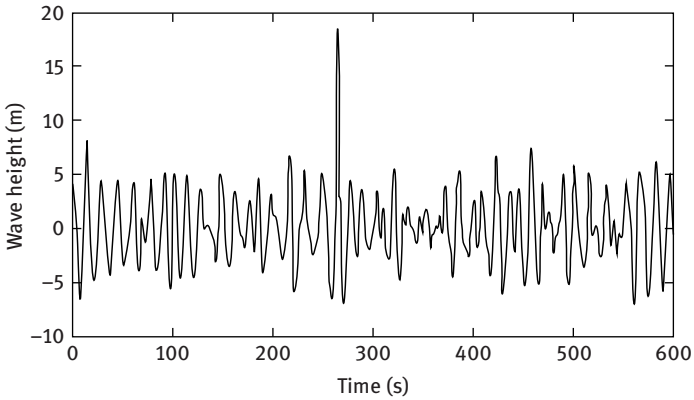


**Figure 1.5:** The New Year wave in the North Sea.

During 1969–1994, 22 supercarriers have been attacked by the rogue waves in the Pacific and the Atlantic Oceans [6], resulting in 525 deaths (for more details see [6]). In addition, according to the available statistics, over the two decades from 1981–2000, destruction of more than 200 supercarriers, because of the heavy sea states and severe weather conditions [7], has been reported.

On January 1, 1995, a rogue wave called "New Year wave" attacked the Draupner platform, which was located in the North Sea which had 70 m depth [8]. The New Year wave was unambiguously recorded, with a peak elevation of 18.5 m and average period of 12.5 s. The distribution of the wave height is shown in Figure 1.5.

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**Figure 1.6:** Rogue wave on the South California coast.

On April 16, 2005, at the Georgia coast, there had been a number of huge waves with at least 7-story height. The waves collapsed like a bow, broke the windows of the ship, fell down into the deck with 10 m height, submerged 62 cabins and injured 4 passengers.

US National Oceanic and Atmospheric Administration Web site published the image of a rogue wave with 18 m height, as seen in Figure 1.6, which was shot from a warship in a few seconds after the striking, in the stormy condition. From the analysis of the above figures and events, we observe the following characteristics of rogue waves: (i) it has a very steep peak; (ii) it is highly asymmetric; and (iii) it appears from nowhere and disappears without a trace.

Humans have long suspected the existence of rogue waves, which was considered as mysterious as the emergence of mermaids and sea monsters. The existence of the rogue wave in the real world was confirmed by direct observations. In the sixties and seventies of the twentieth century, oceanographers began to believe in the existence of rogue waves. Radar satellites were launched in 1991 and 1995, by the scientists from the European Space Agency, the German Aerospace Center and several other European research institutions. The radar beam is seen as potentially the ideal tool for measuring wave height, based on the time for the radar beam to return from the sea. On January 1, 1995, the Draupner oil platform in the North Sea was hit by a wave with a maximum height of 84 ft, which was measured by an onboard laser device. It was the first scientific evidence for the existence of the rogue wave. In February 2000, the British ocean survey ship measured a giant wave with 95 ft height in the west of Scotland. Geophysical researchers say this is the biggest rogue wave which has been ever recorded by the scientific instruments. Studies have shown that the height of rogue wave rises up to at least 82 ft, about the height of 8 stories. The average height of the huge rogue wave is about 100 ft. The European MaxWave Project was conducted in Brest in 2000; its main goal was to improve the understanding of the rogue wave by using the

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new oceanographic knowledge and new techniques with radar satellites [7]. In 2005, Muller et al. in the "Rogue Waves—The Fourteenth Aha Hulikoa Hawaiian Winter Workshop" reported that "Our understanding of rogue waves is greatly hampered by the lack of comprehensive observations in space and time." [9] Undoubtedly, comprehensive, long-term measurements of the rogue wave are very necessary and urgently needed.

Maritime observers have named the rogue wave as "wall of water," "sea cave," or consecutive high waves. This type of wave appears in a peaceful environment without any warning. The outstanding peak is surrounded by deep holes. Corresponding to the large-amplitude wave, rogue wave is seemed as an isolated huge wave with amplitude much larger than the average wave peaks in the surrounding area.

Due to limitations of observing conditions, people had not yet fully understood in this phenomenon. Scientists have not reached a consensus on the rogue wave, and have neither given its definition, nor calculated the probability of occurrence. Although there is no precise definition of rogue wave, it should include the following characteristics: high waves, very asymmetrical, very steep. Its amplitude should at least be twice the amplitude of the background field (significant). The large-amplitude wave which appears in the ocean suddenly, and the huge wave whose amplitude is twice the amplitude of the surrounded average peaks, are usually called rogue waves. The current understanding of the rogue wave is preliminary and superficial.

Rogue waves can be considered as a new "nonlinear scientific revolution" in nonlinear science, after soliton [10–12]. From the mathematical point of view, there are fundamental differences between rogue wave and soliton solutions. With the evolution of time, soliton can propagate with a stable envelop. Even for some nonautonomous system, the nonautonomous solitons almost have the similar shapes. However, the rogue wave is unstable and sensitive with the initial data, its shape is variable with the evolution of time.

## 1.2 Some Famous Experiments of Rogue Waves

In 2007, Solli et al. reported the observation of "optical rogue waves" in nonlinear optics for the first time in *Nature* [13]. The propagation of the optical beams can be governed by the generalized nonlinear Schrödinger (NLS) equation

$$\frac{\partial A}{\partial z} - \mathrm{i}\frac{\mathrm{i}^m \beta_m}{m!}\frac{\partial^m A}{\partial t^m} = \mathrm{i}\gamma\left[|A|^2 A + \frac{\mathrm{i}}{\omega_0}\frac{\partial}{\partial t}(|A|^2 A) - T_g A\frac{\partial |A|^2}{\partial t}\right], \tag{1.2.1}$$

where $\beta_m$ is the optical fiber dispersion coefficient, $\gamma$ is the nonlinear coefficient of the fiber, $\omega_0$ is the central carrier frequency of the field, $T_g$ is the parameter that characterizes the delayed nonlinear response of silica fiber. $A = A(t, z)$ is the slowly varying electric field envelope, $z$ means the variable, $t$ denotes the time, $\frac{\partial}{\partial t}(|A|^2 A)$ is

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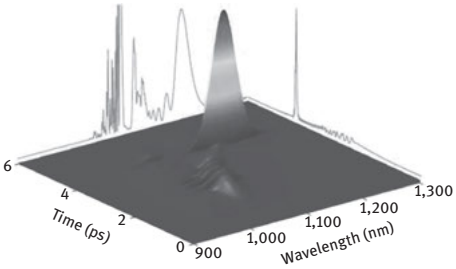


**Figure 1.7:** The profile of the optical rogue wave.

the self-steepening, $A\frac{\partial}{\partial t}(|A|^2)$ represents the vibrational Raman response of the medium, $|A|^2A$ stands for Kerr nonlinearity, $\frac{\partial^m A}{\partial t^m}$ means the dispersion term. Based on the supercontinuum generation in the presence of noise, the optical rogue wave was observed in the experiments, as seen in Figure 1.7.

In 2010, the Peregrine soliton (rogue wave) was first observed in an experiment, which is in very good agreement with the numerical simulations and Peregrine's analytic prediction [14]. These results were reported in *Nature Physics* by Kibler et al. The physical model is governed by the well-known NLS equation

$$\mathrm{i}\frac{\partial\psi}{\partial\xi} + \frac{1}{2}\frac{\partial\psi}{\partial\tau^2} + |\psi|^2\psi = 0, \tag{1.2.2}$$

where $\tau$ and $\xi$ denote the time and space, respectively. Equation (1.2.2) has the solution in the following form:

$$\psi(\xi,\tau) = \left[1 - \frac{4(1+2\,\mathrm{i}\xi)}{1+4\tau^2+4\xi^2}\right]e^{\mathrm{i}\xi}.$$

Figure 1.8 shows us the correlation among the experiment observation, numerical simulations and analytic solution.

Here we present some fresh results on models of the deep ocean, although there are lots of studies on it. Since it is very hard and dangerous to do experiments in the deep ocean, Chabchoub et al. observed rogue wave in a water wave tank [15] in 2011. In turn, in 2012, Chabchoub, et al. observed a hierarchy of up to fifth-order rogue wave in a water tank [16]. Thus, these experiments confirmed the existence of the rational rogue wave solutions. In addition, the characteristics rational rogue wave solution are similar to that of rogue wave. So in the following chapters, we define the rational function solution as the rogue wave solution.

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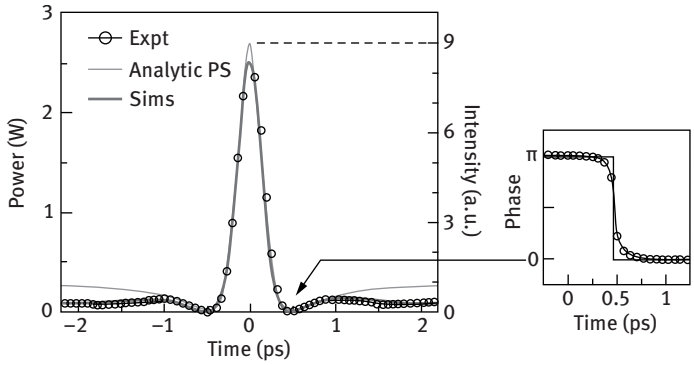


**Figure 1.8:** The correlation between the experiment observation, numerical simulations and analytic solution.

The deepwater wave model can be described by the following NLS equation:

$$\mathrm{i}\left(\frac{\partial a}{\partial t} + c_g \frac{\partial a}{\partial x}\right) - \frac{\omega_0}{8k_0^2}\frac{\partial^2 a}{\partial x^2} - \frac{\omega_0 k_0^2}{2}|a|^2 a = 0, \tag{1.2.3}$$

which was first derived by Zakharov; $a(x,t)$ is related to the surface elevation, where $t$ and $x$ denote time and spatial coordinates, respectively, $k_0$ and $\omega_0 = \omega(k_0)$ are the wave number and wave frequency of the carrier wave, $\omega_0 = \sqrt{gk_0}$, $g$ is the gravitational acceleration, $c_g = \frac{d\omega}{dk}|_{k=k_0} = \frac{\omega_0}{2k_0}$ is the group velocity. Through the scale transformations $T = -\frac{\omega_0}{8k_0^2}t$, $X = x - c_g t = x - \frac{\omega_0}{2k_0}t$ and $q = \sqrt{2}k_0^2 a$, we obtain the following dimensionless form of NLS equation:

$$\mathrm{i}q_T + q_{XX} + 2|q|^2 q = 0. \tag{1.2.4}$$

The first rogue wave solution for Eq. (1.2.4) reads as

$$q(X,T) = \left(1 - \frac{4(1+4\mathrm{i}T)}{1+4X^2+16T^2}\right)\mathrm{e}^{2\mathrm{i}T}.$$

To describe the experiment results, we introduce the transformations $X \to \sqrt{2}k_0^2 a_0 (x - c_g t)$, $T \to -\frac{k_0^2 a_0^2 \omega_0}{4}t$. Thus, the the first rogue wave solution $q(X,T)$ can be transformed into

$$q(x,t) = a_0 \mathrm{e}^{-\frac{\mathrm{i}k_0^2 a_0^2 \omega_0}{2}t}\left(1 - \frac{4(1-\mathrm{i}k_0^2 a_0^2 \omega_0 t)}{1+[2\sqrt{2}k_0^2 a_0 (x-c_g t)]^2 + k_0^4 a_0^4 \omega_0^2 t^2}\right).$$

The comparison between the analytic solution and experimental data is shown in Figure 1.9, which confirmed the existence of rogue wave solution in water models.

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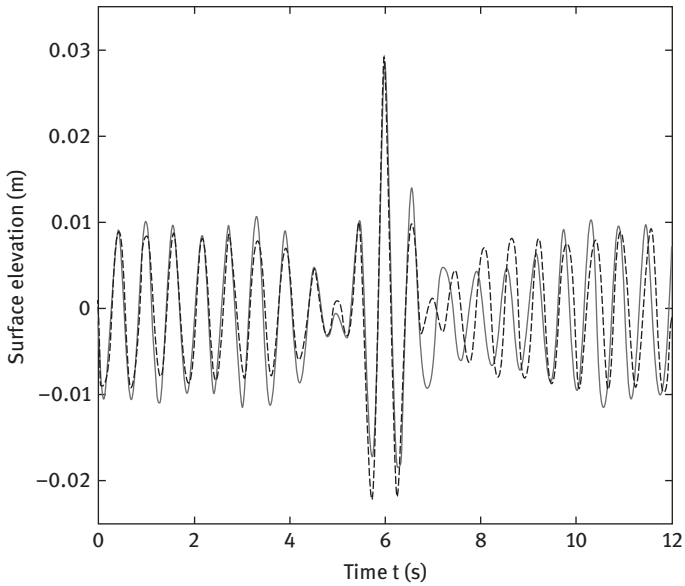


**Figure 1.9:** The comparison between the analytic solution and experimental data at $X = 0$ with $a_0 = 0.01\,\mathrm{m}$, $k_0 = 11.63\,\mathrm{m}^{-1}$, $\omega_0 = 10.7\,\mathrm{s}^{-1}$.

The high-order rogue wave solution will be discussed in detail in the following chapters, due to its complex expression. Figure 1.10 displays the fifth-order rogue wave and experiment observation, respectively.

Among many viewpoints and hypotheses on the rogue wave, our basic assumptions are that the rogue wave can be formed via modulation instability and the rational rogue wave solution is in accordance with the rogue wave. Graphically, the rational solution which we obtained almost coincides with the characteristics of rogue wave. But another inconsistent characteristic of rogue wave is that it is highly asymmetric, which is inconsistent with the strange symmetry of the first-order rational rogue wave solution. However, this is not the sticking point, since the rogue wave phenomenon is very complicated. All in all, the above experiments confirmed that our viewpoints are reasonable.

The analysis on the rogue wave solution for some integrable models is the aim of this book. In theory, integrable model is very beautiful in mathematical expression, but with less physical characters. One advantage of integrable models is that we can find some analytical solutions for them. There are several methods to solve the integrable systems, including the inverse scatting transformation (IST), algebro-geometric approach, Darboux transformation and Hirota bilinear method. The IST is consistent with Darboux transformation method in essence. Darboux transformation is more convenient in the aspect of constructing the exact solution. Algebro-geometric approach relies on constructing the Baker analytic functions and spectral analysis.

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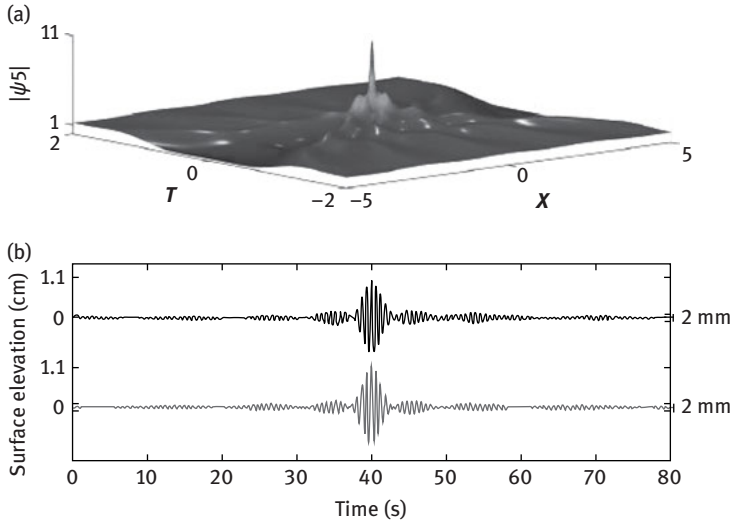


**Figure 1.10:** (a) Fifth-order rogue wave; (b) Experimental demonstration of the fifth-order rogue wave at $x = 0$ with $a_0 = 1\,\text{mm}$, $\epsilon = a_0 k_0 = 0.01$ (upper curve) compared to the theoretical analytic solution at the same position (down curve).

Once an integrable model has been bilinearized through certain transformation, with the truncated parameter expansion at different levels, a series of solutions can be obtained through the bilinear method. The mentioned methods can also be applied to get the rogue wave solution with some modifications. In Chapter 2, we focus on the generalized Darboux transformation, which was first proposed by Guo, Ling and Liu. The Hirota bilinear method, algebro-geometric approach and inverse scattering method are discussed in Chapter 3. In Chapter 4, we present the rogue wave in nonautonomous nonlinear systems by using the similarity transformation and parameter analysis, which are the effective ways to deal with the models with variable coefficients.

For the benefit of looking for other viewpoints of rogue wave conveniently, we list some references objectively to let the reader see the recent progresses on rogue wave.

## 1.3 Research Method and Physical Mechanism of Rogue Waves

### 1.3.1 Methodology of Rogue Waves

In the past three decades, the development of rogue wave research is so fast that various physical models which can describe the rogue wave phenomenon were proposed. The main methods are described below.

(1) Models built by experiments and numerical stimulations.

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In 2001, M. Onorato et al. [17] studied the generation in a random sea state. They displayed the extensive numerical simulations of the NLS equation, how the rogue waves in a random sea state are more likely to occur for large values of the Phillips parameters $\alpha$ and the enhancement coefficient $\gamma$.

In 2002, A. Calini and C. M. Schober [18] numerically investigated the dispersive perturbations of the NLS equation, which describes the wave dynamics in deepwater. In addition, they observed that the likelihood of the rogue wave formation can be greatly increased in a chaotic regime.

In 2004, M. A. Tayfun et al. [19] described that the statistics of the largest crest in $N$ waves are used to examine the nature of two exceptionally large waves recorded within the two hourly measurements. Although these waves possess similar characteristics with that of rogue waves, they are shown to be the simple outliers, which may be predicted well within the context of the extreme-value statistics.

In 2009, experiment results made by X. Z. Zhao et al. [20] showed that the possibility occurrence of rogue wave in deepwater is much higher than in shallow water.

In 2011, clear evidence of rogue wave in a multi-stable system was revealed by A. N. Pisarchik et al. [21] through experiments with an erbium-doped fiber laser driven by harmonic pump modulation. They pointed out that the formation of rogue wave depends on the interplay of stochastic processes with multi-stable deterministic dynamics.

(2) Rogue wave can be described by solutions for the nonlinear partial differential equations, i.e., NLS equation, Korteweg–de Vries (KdV) equation, Kadomtsev–Petviashvili (KP) equation, Ablowitz–Ladik (AL) equation, Hirota equation, Davey–Stewartson system, Zakharov system, etc.

In 2009, A. Ankiewicz et al. [22] studied the effect of various perturbations on rogue wave to test whether rogue waves are against the perturbations.

In 2010, the integral relations of the rogue wave solutions for the focusing NLS equation were given by A. Ankiewicz et al. [23].

In 2011, N. Akhmediev et al. [24] studied the spectra of the Peregrine soliton and higher-order rational solutions for NLS equation, which they used as a model in optics and deep ocean. These solutions have the specific triangular spectra which can be easily measurable in optical systems and obey the characteristics of ocean environment. The possibility of early detection and possible localized warning of the appearance of rogue wave are increased, since the triangular feature of the solutions happened at the early stage of their evolution. The “early warning spectra” of rogue waves may become an important research field in the future [24].

### 1.3.2 Physical Mechanism of Rogue Waves

Physical occurrence mechanism of rogue wave is one of the research focuses. Till now, the main mechanisms which are used to describe the occurrence of rogue wave are described below.

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(1) Interaction between water waves.
In 2003, P. Peterson et al. [25] analyzed the interaction between two long-crested shallow water waves in the framework of two-soliton solutions for KP equation. The results showed that the extreme surface elevations are four times the amplitudes of the incoming waves.
(2) Three-dimensional (3D) directional wave focusing.
In 2007, C. Fochesato et al. [26] deemed that the 3D directional wave focusing contributes to the generation of rogue wave in the ocean, which was simulated and analyzed in a 3D numerical wave tank.
(3) Nonlinear focusing of transient frequency modulation wave group.
In 2008, E. Pelinovsky et al. [27] discussed the kinds of ships of rogue waves in the nonlinear focusing mechanism framework of transient frequency modulation wave group.
(4) Modulation instability.
In 2010, M. Shats et al. [28] reported the first observation of extreme wave events in parametrically driven capillary waves. The evidence of strong four-wave coupling in nonlinear waves was first present. And they pointed out that modulation instability is the main mechanism of the occurrence of rogue wave.
(5) Dispersive focusing of unidirectional wave packets.
In 2011, one of the possible mechanisms of the occurrence of rogue wave in deep water was analyzed by E. Pelinovsky et al. [29], based on the dispersive focusing of unidirectional wave packets. This mechanism is associated with the frequency dispersion of water waves and reflected in the interference of many spectral components which are moving with different velocities. Under typical conditions, the characteristic lifetime of an abnormal wave in the framework of this mechanism is approximately two minutes. During its lifetime, a rogue wave changes its shape from a high ridge to a deep depression.
(6) Dispersion enhancement of transient wave groups, geometrical focusing in basins of variable depth and wave-current interaction [30].
(7) Effect of wind.
On the one hand, the observation shows that rogue wave may occur in the presence of wind; on the other hand, focused and unfocused phase asymmetric behavior has been found based on the dispersive focusing mechanism of rogue wave formation under calm and windy conditions. Jeffreys sheltering theory played a significant role in the coherence of anomalous wave group. Numerical simulation verified that the lifetime of rogue wave increases with the higher wind speed [31].
(8) Randomly united by some solitary waves.
(9) Soliton interaction with energy exchanging.

In recent years, with the developments of researches on both theory studies and experiment observations, rogue wave phenomena have been observed in optical pulses, BEC, superfluid, atmospherics and finance, besides in the ocean.

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In 2010, G. Genty et al. [32] discussed the generation process of optical rogue wave in terms of interaction and turbulence. Rogue soliton can be generated from the third-order dispersion or Raman scattering independently, which was concluded from the simulations of picosecond pulse propagation in optical fibers. Simulations of rogue soliton emergence with dispersive perturbation in the long-distance limit were also presented [32].

In 2010, L. Stenflo and M. Marklund [33] showed the possibility of the existence of rogue wave in atmosphere.

In 2010, Z. Y. Yan [34] performed the financial rogue waves in a Ivancevic option pricing model, which can replace the Black–Scholes model. These rogue wave solutions can depict the possible physical mechanisms for rogue wave phenomenon in financial markets and related fields.

There exist some problems in the studying of the rogue wave: (1) Formation mechanism and characters of rogue wave; (2) Measurement of rogue wave; (3) New discovery of rogue wave models; (4) Applications of rogue wave; (5) Experiment excitation of rogue wave; (6) Prediction of rogue wave.

With the development of science and technology, real-time dynamics monitoring carried by remote-sensing technology has provided massive reliable information of rogue wave. Large number of marine disasters prompted researchers to study the rogue wave. Applications of the rogue wave studies are the prediction of the occurrence of rogue wave based on the rogue wave frequency, marine and ship design, and prevention of economic crisis. In a word, deep understanding of rogue waves not only is good for the ship design, but also has a great significance on the marine safety once we can predict it as weather forecast.

## 1.4 Mechanisms of Rogue Waves

In this chapter, we cite some reports of the mechanisms of rogue wave which are discussed in detail in Ref. [6].

### 1.4.1 Linear Mechanisms of Rogue Waves

#### 1.4.1.1 Dispersion Enhancement of Transient Wave Groups (Spatio-Temporal Focusing)

If a short wave with small group velocity is located in front of a long wave which possesses large group velocity, long wave will overtake the short wave with the time development. During the interaction process, a large-amplitude wave will appear at some fixed time owing to the nonlinear superposition. Afterward, the long wave will be in front of the short wave, and the amplitude of large-amplitude wave train will decrease. A significant focusing of the wave energy can occur when all the quasi-monochromatic groups merge at a fixed location. Such an initial specific location of

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transient wave groups may lead to the formation of rogue wave with the resonant dynamics of the wave generation by the increasing wind. This evolution can explain the reason of the short lifetime of rogue wave. To illustrate the dispersive focusing of unidirectional water waves quantitatively, we take the kinematic equation for characteristic wave frequency $\omega$ into account

$$\frac{\partial \omega}{\partial t} + c_{gr}(\omega)\frac{\partial \omega}{\partial x} = 0, \tag{1.4.1}$$

where the group velocity $c_{gr}(\omega) = d\omega/dk$ can be calculated via the dispersion relation

$$\omega = \sqrt{gk\ \tanh(kh)}, \tag{1.4.2}$$

with $h$ being the water depth, $g$ being the acceleration of gravity. For convenience, we set $h$ as a constant. Multiply $\frac{dc_{gr}}{d\omega}$ to both sides of Eq. (1.4.1), we get

$$\frac{\partial c_{gr}}{\partial t} + c_{gr}\frac{\partial c_{gr}}{\partial x} = 0. \tag{1.4.3}$$

The physical explanation of Eq. (1.4.3) is that each component of spectral wave propagates with its own group velocity. The corresponding Riemann wave solution for Eq. (1.4.3) is

$$c_{gr}(x, t) = c_0(\xi) = c_0(x - c_{gr}t), \tag{1.4.4}$$

where, $c_0(x)$ means the initial distribution of waves with different frequencies in space. The form of such a kinematic wave is continuously varied with distance (time), and its slope is calculated from Eq. (1.4.4)

$$\frac{\partial c_{gr}}{\partial x} = \frac{\frac{dc_0}{d\xi}}{1 + t\frac{dc_0}{d\xi}}. \tag{1.4.5}$$

$\frac{dc_0}{d\xi} < 0$ (or $\frac{dc_0}{dx} < 0$ at $t = 0$) stands for the case of long waves behind the short ones, and the initial increase of the slope of the kinematic wave up to infinity with following decrease, which corresponds to the long waves overtaking the short ones during this process. Several waves with different frequencies mixed together at $t = \frac{1}{max(-dc_0/dx)}$. It is obvious that several focusing points are possible for arbitrary transient waves. The self-similar solution of Eq. (1.4.3) can describe the case of all waves meeting at the same point $(x_0, T_f)$,

$$c_{gr} = \frac{x - x_0}{t - T_f}. \tag{1.4.6}$$

Neglecting the capillary effects, the zone of the variable wave group compresses from $(gh)^{1/2}T_f$ to zero at the fixed time $T_f$, once the group velocity of the water waves varies

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from $(gh)^{1/2}$ to zero. From the solution (1.4.6), we can see that the corresponding variation of the wave frequency is required for optimal focusing. For example, for the deepwater case ($c_{gr} \sim \frac{1}{\omega}$), a wave train with a variable frequency $\omega \sim (t_0 - t)$, which could be generated by the paddle in the laboratory tank, is necessary to provide the maximum effect (optimal focusing). The wave amplitude satisfies the energy conservation equation

$$\frac{\partial A^2}{\partial t} + \frac{\partial (c_{gr} A^2)}{\partial x} = 0, \tag{1.4.7}$$

whose solution can be expressed explicitly as

$$A(x,t) = \frac{A_0(x)}{\sqrt{1 + t\,\frac{dc_0}{d\xi}}},$$

where $A_0(x)$ is an initial distribution of wave amplitude in space. The wave has infinite amplitude at each focal point, while near the focal point, the amplitude $A \sim (T_f - t)^{-1/2}$.

The probability of the occurrence of rogue wave should be very high, since the wind waves are seemed as the frequency and amplitude-modulated wave groups, and the infinite wave height at caustics point is predicted by the kinematic approach. But in fact, the actual situation is much more complex. Kinematic approach assumes slow variations of the amplitude and frequency (group velocity) along the wave group; however, the approximation is not valid in the vicinity of the focal points. It is a well-known problem in the ray methods, not only for water waves. Various expressions of the Fourier integral for the wave field near the caustics made up the generalizations of the kinematic approach in linear theory, which was expressed through the Maslov integral representation. Here, we consider the simplified form of such a representation for conditions of optimal focusing (1.4.6) and use the standard form of the direct and inverse Fourier transformation for water wave displacement,

$$\eta(x,t) = \int_{-\infty}^{+\infty} \eta(k) e^{i(kx - \omega t)} dk,$$
$$\eta(k) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \eta_0(x) e^{-ikx} dx,$$

where $\eta_0(x) = \eta(x,0)$ is the initial water displacement in unidirectional wave field, $\omega$ is the wave frequency which satisfies Eq. (1.4.2). $\eta(x,t)$ can be derived via the delta-function and stationary-phase method,

$$\eta(x,t) = Q\sqrt{\frac{c_{gr}}{2\pi x\,|d\,c_{gr}/dk|}}\, e^{i(kx - \omega t - \frac{\pi}{4})}, \tag{1.4.8}$$

where $c_{gr}$ is calculated from the conditions for optimal focusing (1.4.6), $Q$ is the intensity of the delta function. At $k \to 0$, Eq. (1.4.8) should be replaced by the following formula:

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$$\eta(x,t) = Q\left(\frac{2}{cth^2}\right)^{\frac{1}{3}} Ai\left[\left(\frac{2}{cth^2}\right)^{\frac{1}{3}}(x-ct)\right], \tag{1.4.9}$$

with $\omega = \sqrt{gh}\left(1-\frac{k^2h^2}{6}\right)$, $Ai(z)$ being Airy function. As a consequence, the amplitude of the leading wave decreases as the rate of $t^{-\frac{1}{3}}$ and the wave length increases as $t^{\frac{1}{3}}$.

#### 1.4.1.2 Spatial (Geometrical) Focusing of Water Waves

Considering two horizontal coordinates $x$ and $y$, wave frequency and wave vector should satisfy the following kinematic equations:

$$\frac{\partial \mathbf{k}}{\partial t} + \nabla\omega = 0, \qquad \nabla \times \mathbf{k} = 0, \tag{1.4.10}$$

where

$$\omega = -\frac{\partial\theta}{\partial t}, \qquad \mathbf{k} = \nabla\theta, \tag{1.4.11}$$

with $\theta$ being the phase of quasi-monochromatic wave $\eta = A(x,y,t)e^{i\theta(x,y,t)}$. The characteristic forms of the above equations are written as

$$\frac{d\mathbf{r}}{dt} = \frac{\partial\omega}{\partial\mathbf{k}}, \qquad \frac{d\mathbf{k}}{dt} = -\frac{\partial\omega}{\partial\mathbf{r}}, \tag{1.4.12}$$

where the wave frequency $\omega$ satisfies the dispersion relation (1.4.2), $r = (x,y)$. Eq. (1.4.12) is a well-known equation of the ray theory written in Hamiltonian form.

Wave amplitude can be given by the two-dimensional version of energy balance equation

$$\frac{\partial A^2}{\partial t} + \nabla\cdot(\mathbf{c}_{gr}A^2) = 0, \tag{1.4.13}$$

which may transform into the energy flux conservation along the ray tube

$$\mathbf{c}_{gr}\,\Lambda\,A^2 = constant,$$

with $\Lambda$ being the differential width of the ray defined as the distance between the adjacent rays. Wave amplitude becomes infinite because $\Lambda$ equals zero at any focal point. The energy balance equation will not be valid in the vicinity of the caustics. Detailed description of the wave field in the caustics vicinity can be done through the asymptotic Maslov representation or exact solutions for some test cases. For example, if $h = h(x)$, the shallow water wave can be illustrated by the ordinary differential equation

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$$g\frac{d}{dx}\left[h(x)\frac{d\eta}{dx}\right] + [\omega^2 - gh(x)k_y^2]\eta = 0, \tag{1.4.14}$$

where $\omega$ stands for the frequency of the monochromatic wave, and $k_y$ means the wave number in $y$-direction. Location of caustics can be obtained with the second bracket of Eq. (1.4.14) being zero. In the neighborhood of caustics, depth can be simplified as $h(x) = h_c(1 + \frac{x}{L})$ with $h = h_c$ at $x = 0$.

Therefore, in the neighborhood of caustics, Eq. (1.4.14) has the Airy equation form

$$\frac{d^2\eta}{dx^2} - \frac{k_y^2}{L}x\eta = 0,$$

whose solution can be described by the Airy function

$$\eta(x) = C \cdot Ai\left(-\frac{x\,k_y^{\frac{2}{3}}}{L^{\frac{1}{3}}}\right), \tag{1.4.15}$$

where $C$ stands for a constant and can be determined through the amplitude of the incident wave $A_0$ by using asymptotic expression for the Airy function far from the caustics. The wave field is bounded on the caustics. As a result, the wave amplification on the caustics is

$$\frac{A_c}{A_0} \sim (Lk_y)^{\frac{1}{6}}, \tag{1.4.16}$$

and it is relatively weak for long waves. We need to point out that the spatial focusing procedure involves not only a wave amplification, but also a change of the wave shape.

#### 1.4.1.3 Wave-Current Interaction

The problem of the wave-current interaction requires a special investigation, since rogue waves were observed frequently in such strong currents like Gulf Stream and Agulhas Current. Formally, the ray pattern is described by the system (1.4.13) while the dispersion relation should be corrected. Considering the deepwater waves case, the dispersion relation for waves on a steady current becomes anisotropic,

$$\omega = \Omega(k) + \mathbf{k}\,U(x, y), \qquad \Omega = \pm\sqrt{gk}.$$

With $U_x(x)$ only, the wave-current interaction is not trivial in one-dimensional case. When the current is opposite to incident monochromatic wave, it blocks the wave at the point $x_0$, where the group velocity is zero,

$$c_{gr} = \frac{d\omega}{dk} = \frac{1}{2}\sqrt{\frac{g}{k}} + U(x_0) = 0. \tag{1.4.17}$$

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The wave phase and group velocities have the same sign when the wave approaches the blocking point; after the reflection from the blocking point, they transform to the opposite sign. In the process of interaction, the wave number increases and an initial long wave transforms to a short wave. Wave amplitude can be found from the wave action balance equation

$$\frac{\partial}{\partial t}\left(\frac{A^2}{\Omega}\right) + \nabla \cdot \left(\frac{c_{gr}A^2}{\Omega}\right) = 0, \tag{1.4.18}$$

which is the generalization of the energy balance equation (1.4.13). For steady currents, Eq. (1.4.18) transforms into the wave action flux,

$$\frac{c_{gr}\Lambda A^2}{\Omega} = constant,$$

with $\Lambda$ being the same as above mentioned. For the propagation of unidirectional wave, the blocking point with zero group velocity (1.4.17) plays the role of caustics and the wave amplitude formally tends to infinity. In fact, Eq. (1.4.18) is not valid in the neighborhood of the caustics. More accurate asymptotic analysis should be calculated by the Maslov integral

$$\eta(x) = C \cdot Ai\left[\left(\frac{8\frac{\partial U}{\partial x}}{\Omega(k_*)}\right)^{\frac{1}{3}} k_*(x - x_0)\right] e^{ik_* x - \omega t},$$

with $k_*$ being the value of the wave number at the blocking point determined by Eq. (1.4.17), and $\frac{\partial U}{\partial x}$ calculated at the same point. As a consequence, wave amplitude at the blocking point is bounded; comparing with Eq. (1.4.16),

$$\frac{A_c}{A_0} \sim \left(\frac{\Omega}{\frac{dU}{dx}}\right)^{\frac{1}{6}}.$$

### 1.4.2 Nonlinear Mechanisms of Rogue Waves

Nonlinear mechanisms of the rogue wave will be analyzed here in the weakly nonlinear limit. In theory, the description of nonlinear mechanisms is more perfect than the linear mechanisms.

#### 1.4.2.1 Weakly Nonlinear "Rogue Wave Packets in Deep and Intermediate Depths

The simplified nonlinear model of 1+1 quasi-periodic deepwater wave trains in the lowest order in wave steepness and spectral width is based on NLS equation

$$\mathrm{i}\left(\frac{\partial A}{\partial t} + c_{gr}\frac{\partial A}{\partial x}\right) = \frac{\omega_0}{8k_0^2}\frac{\partial^2 A}{\partial x^2} + \frac{\omega_0 k_0^2}{2}|A|^2 A, \tag{1.4.19}$$

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where the surface elevation $\eta(x,t)$ is given by

$$\eta(x,t) = \frac{1}{2}\left(A(x,t)e^{\mathrm{i}(k_0x-\omega_0t)} + c.c. + \cdots\right), \tag{1.4.20}$$

with $k_0$ and $\omega_0$ being the wave number and frequency of the carrier wave, respectively, *c.c.* being the complex conjugate, and "$\cdots$" being the weak highest harmonics of the carrier wave. The complex wave amplitude $A$ is a slowly varying function of $x$ and $t$.

NLS equation plays an important role in the understanding of nonlinear dynamics of water waves. It is well known that a uniform train of amplitude $A_0$ is unstable to the modulation instability of the long disturbances of wave number $\Delta k$, i.e.,

$$\frac{\Delta k}{k_0} < 2\sqrt{2}k_0A_0. \tag{1.4.21}$$

The maximum instability occurs at $\frac{\Delta k}{k_0} = 2k_0A_0$ with the maximum growth rate that is equal to $\frac{\omega_0(k_0A_0)^2}{2}$. The nonlinear stage of the modulation instability has been deeply investigated analytically, numerically and experimentally. In fact, the rogue wave solutions for NLS equation can describe the rogue wave phenomenon in deepwater properly.

For 3D wave trains, the governed (2 + 1)-D NLS equation is

$$\mathrm{i}\left(\frac{\partial A}{\partial t} + c_{gr}\frac{\partial A}{\partial x}\right) = \frac{\omega_0}{8k_0^2}\frac{\partial^2 A}{\partial x^2} - \frac{\omega_0}{4k_0^2}\frac{\partial^2 A}{\partial y^2} + \frac{\omega_0k_0^2}{2}|A|^2A. \tag{1.4.22}$$

The above 2 + 1-D NLS equation is principally anisotropic, and modulations of wave packets in the longitudinal and transversal directions behave differently. Specially, modulations in the transverse direction are stable. The domain of the modulation instability is found as

$$\frac{\sqrt{2}\Delta k_y}{k_0} < \frac{\Delta k_x}{k_0} < \sqrt{\frac{2(\Delta k_y)^2}{k_0^2} + 8k_0^2A_0^2},$$

where $\Delta k_x$ and $\Delta k_y$ stand for the wave numbers in $x$ and $y$ directions, respectively. The similar rogue wave phenomenon will be observed through the numerical simulation in the domain of modulation instability.

3D wave trains in the finite depth water can be described by the Davey–Stewartson system in dimensionless form

$$\mathrm{i}\frac{\partial q}{\partial t} + \frac{\partial^2 q}{\partial x^2} - \frac{\partial^2 q}{\partial y^2} + 2q(|q|^2 - p) = 0, \tag{1.4.23a}$$

$$s_1\frac{\partial^2 p}{\partial x^2} + s_2\frac{\partial^2 p}{\partial y^2} = \frac{\partial^2 |q|^2}{\partial x^2}, \tag{1.4.23b}$$

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with

$$s_1 = S_1\frac{N_1}{N_2^2},\quad s_2 = S_2\frac{N_1}{N_2^2}\frac{D_1}{D_2},\quad D_1 = -\frac{4k}{c_{ph}}\frac{\partial c_{gr}}{\partial k},\quad D_2 = \frac{4c_{gr}}{c_{ph}},$$
$$N_1 = \frac{1}{8}(1+9\sigma^{-2}-13(1-\sigma^2)-2\sigma^4),\quad N_2 = 1+\frac{c_{gr}}{2c_{ph}}(1-\sigma^2),$$
$$S_1 = \frac{kh}{\sigma}-\frac{c_{gr}^2}{c_{ph}^2},\quad S_2 = \frac{kh}{\sigma},\quad \sigma = \tanh(kh),$$

$c_{gr}$ and $c_{ph}$ being the group and phase velocities, respectively. The surface elevation is expressed as

$$\eta(X,Y,T) = \frac{c_{ph}^2}{g\sqrt{2N_1}}[q(x,y,t)e^{\mathrm{i}(\omega T-kX)}+c.c.]+\cdots, \tag{1.4.24}$$
$$x = \frac{2k}{\sqrt{D_1}}(X-c_{gr}T), y = \frac{2k}{\sqrt{D_2}}Y, t = \frac{1}{2}\omega T,$$

where $p$ is the mean flow of the wave envelop, $N_1$ and $N_2$ mean the nonlinear coefficients, $D_1$ and $D_2$ stand for the longitudinal and transverse dispersions, respectively. All the coefficients of Eq. (1.4.23) are related with $kh$ only. With the negligible of the mean flow in deep water, DS system (1.4.23) reduces to the (2 + 1)-D NLS Eq. (1.4.22).

#### 1.4.2.2 Dysthe Equation

We start with the incompressible flow equation with finite depth of $h$ [35, 36],

$$\nabla^2\phi = 0, \qquad -h < z < \zeta; \tag{1.4.25a}$$
$$\frac{\partial^2\phi}{\partial t^2}+g\frac{\partial\phi}{\partial z}+\frac{\partial}{\partial t}(\nabla\phi)^2+\frac{1}{2}\nabla\phi\cdot\nabla(\nabla\phi)^2 = 0, \qquad z = \zeta; \tag{1.4.25b}$$
$$\frac{\partial\zeta}{\partial t}+\nabla\phi\cdot\nabla\zeta = \frac{\partial\phi}{\partial z}, \qquad z = \zeta; \tag{1.4.25c}$$
$$\frac{\partial\phi}{\partial z} = 0, \qquad z = -h. \tag{1.4.25d}$$

where $\phi$ is the velocity, $\zeta$ means the surface elevation, $g$ is the acceleration of gravity, $z$ stands for the vertical coordinate, $(x, y)$ is the location in the horizontal direction, and $\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$. Set

$$ka = O(\epsilon), \frac{|\Delta k|}{k} = O(\epsilon), (kh)^{-1} = O(\epsilon), \tag{1.4.26}$$

and expand the velocity $\phi$ and surface elevation $\zeta$ as

$$\phi = \bar{\phi}+\frac{1}{2}\left(A\,e^{\mathrm{i}\,\theta+kz}+A_2\,e^{2(\mathrm{i}\,\theta+kz)}+\cdots+c.c.\right),$$
$$\zeta = \bar{\zeta}+\frac{1}{2}\left(B\,e^{\mathrm{i}\,\theta}+B_2e^{2\mathrm{i}\,\theta}+\cdots+c.c.\right),$$

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where phase $\theta = kx - \omega t$ is the carrier wave vector along $x$-direction. The slow drift $\bar{\phi}$ and set-down $\bar{\zeta}$ as well as the harmonic amplitudes $A, A_2, \ldots, B, B_2, \ldots$ are the functions of the slow modulation variables $\epsilon x$ and $\epsilon t$. $\bar{\phi}$ depends on the vertical variable $\epsilon z$, and $A, A_2, \cdots$ depend on the basic vertical coordinate $z$.

From the leading-order perturbation problem for the first harmonic, the dispersive relation of the deep gravity wave is derived as

$$\omega^2 = gk.$$

Introducing the coordinate transformations,

$$\begin{aligned}
&\omega t \to t,\\
&k(x,t) \to (x,t),\\
&k(B, B_n, \bar{\zeta}) \to (B, B_n, \bar{\zeta}),\\
&k^2\omega^{-1}(A, A_n, \bar{\phi}) \to (A, A_n, \bar{\phi}),
\end{aligned}$$

and expanding as $\epsilon$ to $O(\epsilon^4)$, following equations will be obtained:

$$\begin{aligned}
&\frac{\partial A}{\partial t} + \frac{1}{2}\frac{\partial A}{\partial x} + \frac{\mathrm{i}}{8}\frac{\partial^2 A}{\partial x^2} - \frac{\mathrm{i}}{4}\frac{\partial^2 A}{\partial y^2} + \frac{\mathrm{i}}{2}|A|^2 A - \frac{1}{16}\frac{\partial^3 A}{\partial x^3} + \frac{3}{8}\frac{\partial^3 A}{\partial x \partial y^2}\\
&+\frac{3}{2}|A|^2\frac{\partial A}{\partial x} - \frac{1}{4}A^2\frac{\partial \bar{A}}{\partial x} + iA\frac{\partial \bar{\phi}}{\partial x} = 0, && z = 0,\\
&\nabla^2\bar{\phi} = 0, && -h < z < 0,\\
&\frac{\partial \bar{\phi}}{\partial z} = \frac{1}{2}\frac{\partial}{\partial x}|A|^2, && z = 0,\\
&\frac{\partial \bar{\phi}}{\partial z} = 0, && z = -h.
\end{aligned}$$

The extended Dysthe equation is presented as

$$\begin{aligned}
&\mathrm{i}\left(\frac{\partial A}{\partial t} + L(\partial_x, \partial_y)A\right) =\\
&\frac{\omega_0 k_0^2}{2}|A|^2 A - \frac{\mathrm{i}\,\omega_0 k_0^2}{4}\left(6|A|^2\frac{\partial A}{\partial x} + A^2\frac{\partial \bar{A}}{\partial x}\right) + k_0 A\frac{\partial \Phi}{\partial x_{z=0}}, && z = 0,\\
&\Delta\Phi = 0, && -\infty < z < 0, \qquad (1.4.27)\\
&\frac{\partial \Phi}{\partial z} = \frac{\omega_0}{2}\frac{\partial |A|^2}{\partial x}, && z = 0,\\
&\frac{\partial \Phi}{\partial z} \to 0, && z \to -\infty,
\end{aligned}$$

with

$$L(\partial_x, \partial_y) = \mathrm{i}\omega_0\left\{\left[\left(1 - \frac{\mathrm{i}}{k_0}\frac{\partial}{\partial x}\right)^2 - \frac{1}{k_0^2}\frac{\partial^2}{\partial y^2}\right]^{\frac{1}{4}} - 1\right\}.$$

Equation (1.4.27) can simulate numerically the New Year wave event properly.

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#### 1.4.2.3 Fully Nonlinear Models of Rogue Waves

Those already mentioned are the weak nonlinear models. In fact, as the huge waves, rogue waves should be considered as strong nonlinear waves. Few studies are on the framework of fully nonlinear and dispersive models. The governing models are the Laplace equation for the velocity potential $\phi(x, y, z, t)$ with fully nonlinear dynamics:

$$\Delta\phi = 0 \qquad -h \le z \le \eta(x, y, t),$$

with the kinematic and dynamic boundary conditions on the sea surface $z = \eta(x, y, t)$

$$\frac{\partial\eta}{\partial t} + \frac{\partial\phi}{\partial x}\frac{\partial\eta}{\partial x} + \frac{\partial\phi}{\partial y}\frac{\partial\eta}{\partial y} - \frac{\partial\phi}{\partial z} = 0,$$
$$\frac{\partial\phi}{\partial t} + \frac{1}{2}|\nabla\phi|^2 + g\eta = 0,$$

and the boundary condition on sea bottom $z = -h$

$$\frac{\partial\phi}{\partial z} = 0.$$

For infinite depth the above condition becomes $\nabla\phi \to 0$, $z \to -\infty$.

## 1.5 Rogue Wave Solutions for Nonlinear Partial Differential Equations

There have been many studies on the rogue wave solutions for nonlinear partial differential equations. The related works become necessary and essential, since the rogue wave phenomena have been observed in various fields. We list some related references here.

In 1999, K. B. Dysthe and K. Trulsen [37] compared the breather solutions for NLS equation with the simulation data. Through which, they tended to strengthen that the Ma breather and the so-called Peregrine solution may provide useful and simple analytical models for rogue wave phenomenon.

In 2000, E. Pelinovsky et al. [38] proposed a method to find the wave trains whose evolution will lead to the formation of rogue wave, based on the spatial-temporal focusing within the framework of the KdV equation. It is shown that rogue waves can be generated not only for specific conditions, but also for relative wide classes of the wave trains.

In 2000, the dynamic behaviors of rogue waves which were described by the solutions of 1 + 1- and 2 + 1-dimensions NLS equations have been investigated by A. R. Osborne et al. [39]. The analytical results for 1 + 1-dimensions and the numerical ubiquitous occurrence of rogue waves and holes in 2 + 1-dimensions have been analyzed.

In 2004, the occurrence probability of rogue waves was predicted by N. Mori [40] via the Edgeworth's form of a cumulative expansion of the probability density function

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of surface elevation. Through the comparison between the statistical theory and field data, they found that rogue wave sometimes cannot be expected by the Rayleigh theory, but can be predicted by the theory they proposed.

In 2005, A. I. Dyachenko and V. E. Zakharov [41] numerically studied the formation of waves of large amplitude (rogue waves) at the surface of the ocean. They observed that rogue waves, similar to Stokes waves, have the same ratio of the wave height to the wave length. The rogue wave will break if it reaches its limiting state. In addition, the physical mechanism of the formation of rogue wave has been discussed.

In 2005, a single weakly 3D large-amplitude wave (rogue wave) on the deep water was obtained based on the numerical experiments [42].

In 2005, A. V. Porubov et al. [43] showed that the generation of the rogue waves in the ocean can be described by the nonlinear two-dimensional shallow water theory, and the simplest two-dimensional long wave model corresponds to the KP equation.

In 2005, A. L. Islas and C. M. Schober [44] correlated the development of rogue waves in oceanic sea states which was characterized by the Joint North Sea Wave Project (JONSWAP) spectrum, with the proximity to homoclinic solutions of the NLS equation, by using the inverse spectral theory of NLS equation. The occurrence and strength of rogue waves can be predicted through the nonlinear spectral analysis.

In 2006, C. M. Schober [45] investigated the rogue waves in deepwater in the framework of NLS equation and modified Dysthe equations. He observed that the likelihood for the formation of rogue wave in a chaotic regime was increased due to the enhanced focusing during the chaotic evolution of the phases.

In 2006, V. P. Ruban [46] reported a breathing rogue wave on deepwater was formed through the numerical simulations of motion of a fully nonlinear equation for long-crested water waves with quasi-random initial condition. This type of breathing rogue wave behaves in an oscillating manner and exists for a relatively long time without significant change of its maximal amplitude.

In 2006, T. Soomere [47] reviewed the results of investigation of soliton interactions of KP equation and their related rogue wave theory. The marine ecosystem in coastal areas may be influenced by the rogue waves.

In 2007, the modulation instability was studied for the down-stream evolution of surface gravity waves by Andonowati et al. [48] based on an explicit solution for NLS equation.

In 2008, V. E. Zakharov [49] found that rogue waves can be identified with the high steepness giant breathers. Zakharov pointed out that the most possible explanation of the giant breather stability is the integrability of Euler equation, which describes a potential flow of deep ideal fluid with free surface. A string of indirected proofs of this integrability has been displayed.

In 2010, J. L. Bona and J. C. Saut [50] investigated several equations without the surface tension: the linear and NLS equations, a class of fractional order Schrödinger

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equation, and the linearized water wave equations. In addition, they commented about how the results may describe the formation of rogue waves in fluid and optical environments.

In 2010, Y. V. Bludov et al. [51] numerically studied the rogue waves in the two-component BEC, which can be described by the coupled Gross–Pitaevskii equations with variable scattering lengths. They pointed out that the existence of rogue wave solutions may be guaranteed by certain combinations of the nonlinear cross interactions coefficients.

In 2010, A. Ankiewicz and N. Akhmediev [52] obtained the rogue wave solutions for the discrete AL and Hirota equations.

In 2010, R. Grimshaw et al. [53] examined the occurrence of rogue wave solutions for Gardner equation, which can be seen as an extension of the KdV equation with quadratic and cubic nonlinearity. Especially, the authors took the Gardner equation with the same sign of the coefficients of cubic nonlinear and linear dispersive terms. Several situations in which rogue wave occurred were identified through the numerical simulations of the evolution of a modulated narrow-band initial wave field.

In 2010, a quasi-rogue wave solution of the Camassa–Holm equation was obtained by Y. Fan and L. X. Tian [54]. The characteristics were similar to rogue waves in shallow water.

In 2010, P. Dubard et al. [55] discussed the multi-rogue wave solutions for NLS equation in hydrodynamics or in nonlinear optics. Some basic properties of multi-position solutions for KdV equation have also been described, which may be linked with the rogue waves.

In 2010, A. Ankiewicz et al. [56] constructed a hierarchy of rational solutions for Hirota equation via the modified Darboux transformation. Numerical simulations showed that the second-order rational solution as a type of rogue wave can be produced from random initial conditions with a given small amplitude of chaotic perturbations.

In 2011, A. Islas and C. M. Schober [57] investigated the dissipative effects on rogue waves based on the one-dimensional Dysthe equation with the nonlinear and linear damping terms. Through the inverse spectral theory of NLS equation, simulations of the damped Dysthe equation of the sea states characterized by JONSWAP spectrum showed that the rogue wave occurred when the JONSWAP initial data were closer to homoclinic data.

In 2011, N. Akhmediev and A. Ankiewicz [58] studied the modulation instability of the discrete constant-background wave of the AL equation. The exact periodic solutions can be considered as a complete resolution of the Fermi–Pasta–Ulam paradox for the AL system.

In 2011, W. M. Moslem et al. [59] studied the dust-acoustic (DA) rogue wave in a dusty plasma composed of negatively charged dust grains, as well as the electrons and ions obeying the $q$-nonextensive distribution functions. The nonlinear evolution of modulationally unstable DA wave packets was predicted by NLS equation.

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In 2011, Yan [60] studied the (2 + 1)-dimensional nonlocal NLS equation. Various types of rogue waves were plotted and analyzed.

In 2011, P. Dubard and V. B. Matveev [61] constructed a multiparametric family of quasi-rational solutions for the focusing NLS equation. These solutions have also been used to construct a large family of smooth, real localized rational solutions for KP-I equation, which are quite different from the multilump solutions first obtained by Bordag et al.

In 2011, the propagation and dynamical behavior of electrostatic surface plasma rogue waves have been discussed by W. M. Moslem et al. [62].

In 2011, D. Laveder et al. [63] reported that rogue waves in the Alfvén wave turbulence regime can be described by the randomly driven derivative NLS equation in the presence of a weak dissipation.

In 2011, P. Gaillard [64] constructed the rogue wave solutions for the focusing NLS equations in terms of $2N$-order Wronskian determinants through the algebro-geometric approach.

In 2011, Wang et al. [65] introduced a new kind of nonautonomous NLS equation with a $t$-dependent potential. In addition, analysis of rogue wave solutions for a special NLS equation with variable coefficients is given.

In 2012, He et al. [66] investigated the rogue waves in resonant erbium-doped fiber system which is governed by the NLS–Maxwell–Bloch equation. The $N$-order breather solutions of the three fields were constructed via Darboux transformation.

In 2012, the propagation and dynamics of ion acoustic rogue wave in white dwarfs and magnetars have been investigated by R. Sabry et al. [67]. The stability of waves with small and long wave number has also been discussed.

In 2012, an explicit analytical representation of the rogue waves for the Fokas–Lenells equation has been given [68], by using the Darboux transformation and Taylor expansion of the associated breather solution.

In 2012, D. J. Kedziora et al. [70] studied the higher-order rational solutions for NLS equation which appear spatiotemporally as triangular arrays of Peregrine solitons, by numerically applying Darboux transformation. The authors demonstrated the way to generate much more mixture of rogue wave solutions.

In 2012, the manipulation of rogue waves in the dispersion decreasing fibers with logarithmic profile and Gaussian profile has been investigated based on the self-similar first- and second-order rogue wave solutions for 3+1-dimensional inhomogeneous NLS equation [71]. The comparison of the propagation behavior of rogue wave in the dispersion decreasing wave guides with logarithmic profile, Gaussian profile and hyperbolic profile has also been displayed.

In 2012, M. Bacha et al. [72] transformed the KdV equation to an NLS equation, which provided that the frequency of carrier wave is much smaller than the ion plasma frequency. They found that the ion-acoustic rogue waves may be drastically affected by electron nonextensivity depending on whether the entropic index $q$ is positive or negative. The results may be useful in understanding the occurrence and basic features of the nonextensive ion-acoustic rogue waves in space plasmas.

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In 2013, A. Ankiewicz et al. [73] presented rogue wave solutions and confirmed their validity through numerical simulations for NLS equation with third-order dispersion, self-steepening and self-frequency shift, whose integrable cases including Sasa–Satsuma and the Hirota equations.

In 2013, Zhao [74] manipulated the nonautonomous rogue waves of BEC by adjusting the time-dependent variable coefficients.

In 2013, B. G. Zhai et al. [75] derived the multi-rogue wave solutions for coupled NLS equation via the modified Darboux transformation.

In 2013, a new multi-rogue wave which exhibits a four-petaled flower structure in temporal-spatial distribution for a three-coupled NLS equation has been obtained by Zhao and Liu [76].

In 2013, A. Pushkarev and V.E. Zakharov [77] reported the numerical detection of a new type of localized structures in the frame of Majda–McLaughlin–Tabak model which can describe the nonlinear gravity waves on the surface of ideal deepwater.

In 2013, dynamics of the generalized rogue wave solutions for DS-II equation have been analyzed by Y. Ohta and J. Yang [78] via the Hirota method.

In 2013, P. Gaillard et al. [79] performed the third-order rational rogue wave solutions with four parameters.

In 2013, Zhang and Hu [80] performed a gauge transformation to derive the exact rogue wave solutions for $(1+1)$-dimension NLS equation with variable coefficients.

In 2013, H. P. Zhu [81] derived the first- and second-order rogue wave solutions in two-dimensional graded-index wave guides for the variable-coefficient NLS equation, with the help of the similarity transformation.

In 2014, S. Chen et al. [82] presented the dark and bright rogue wave solutions for the long–wave–short-wave resonance equations.

In 2014, H. N. Chan et al. [83] calculated the rogue wave solutions for a derivative NLS equation as a long wave limit of a breather. The authors displayed that rogue wave can occur in the regime of negative cubic nonlinearity if a sufficiently strong self-steepening nonlinearity was also present.

In 2014, the analytical bright-dark and dark-dark rogue wave solutions for the vector NLS equations in the defocusing regime have been reported [84]. The relationship between modulation instability and rogue waves has also been displayed.

In 2014, N. V. Priya et al. [85] investigated a coupled generalized NLS equation. They constructed dark-dark soliton in the defocusing regime, while the general breather (GB), Akhmediev breather (AB), Ma soliton (MS) and rogue wave solutions were in the focusing regime. The structures of GB, AB, MS and rogue wave profiles have been analyzed with the four-wave mixing parameters. In addition, they demonstrated the way of capturing rogue wave solutions from AB and MS.

In 2014, soliton solutions and second-order rogue wave solutions for an extended NLS equation with third- and fourth-order terms with variable coefficients have been derived [86].

In 2014, W. R. Sun [87] derived the double-Wronskian solitons and rogue waves solutions for the inhomogeneous NLS equation in an inhomogeneous plasma.

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In 2015, D. W. Zuo et al. [88] presented the higher-order rogue-wave solutions for a generalized NLS–Maxwell–Bloch system via the generalized Darboux transformation. The effects of gain or loss term and the linear and Kerr nonlinearity inhomogeneous parameters on the bright rogue wave $q$, bright-dark rogue wave $p$ and dark rogue wave $\eta$ have been analyzed.

In 2015, N. V. Priya and M. Senthilvelan [89] obtained the exact breather and rogue wave solutions for a coupled generalized NLS equation through Darboux transformation. Evidence for the connection between the occurrence of rogue wave solution and the modulation instability has been given.

In 2015, Z. Y. Yan [90] studied the two-dimensional vector rogue waves and interactions of the two-dimensional inhomogeneous two-component Gross–Pitaevskii equations with varying potentials, through similarity transformations and controlled coefficients.

In 2015, G. Weerasekara et al. [91] numerically analyzed the rogue wave phenomenon associated with the NLS equation with weak third-order dispersion. They pointed out that the quasi-soliton interaction process was the generation mechanism of rogue wave.

## 1.6 Optical Rogue Waves

The "optical rogue wave" has been proposed for a long time, which possesses strong application background. The optical rogue wave can be seen as the extension of the optical soliton. Here we list some results for the convenience of readers.

In 2002, D. Clamond and J. Grue [92] observed that the interaction between envelop solitons may generate large waves, through the studies of the long time evolution of a surface gravity wave packet numerically, based on a simplified model of NLS-like equation.

In 2008, D. R. Solli et al. [93] demonstrated that optical rogue waves with rare and bright flashes of broadband light arising in subthreshold supercontinuum generation are initiated by an exceedingly weak stimulus. Through this effect, the optical ultrastable bright supercontinuum with greatly enhanced coherence has been produced.

In 2008, J. M. Dudley et al. [94] present a numerical study of the evolution dynamics of "optical rogue waves" with statistically rare extreme red-shifted soliton pulses arising from supercontinuum generation in photonic crystal fiber. They demonstrated that the rogue wave generation can be enhanced by an order of magnitude through a small modulation across the input pulse envelope and effectively suppressed by the using of a sliding frequency filter.

In 2009, the evolution of a random directional wave in deepwater was studied in a laboratory wave tank by using a directional wave generator [95]. Therefore, the occurrence of extreme wave in the sea can be seen as a result of the interaction between quasi-resonant waves.

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In 2009, C. Liu [96] presented an adaptive numerical filter for analyzing fiber-length-dependent properties of optical rogue waves. This filter was used to study a data set of 1000 simulated supercontinuum pulses, which was produced from 5ps pump pulses containing random noise.

In 2010, the characteristics of optical rogue waves in supercontinuum generation in the femtosecond regime have been studied experimentally by M. Erkintalo [97].

In 2010, D. R. Solli et al. [98] reviewed the works on optical rogue waves and an ultrafast phenomenon counterpart to the freak ocean waves known to roam the open oceans. The heavy-tailed statistical properties of optical rogue waves have been discussed.

In 2010, G. Gentyand and C. M. Sterke [99] discussed the optical rogue wave generation in terms of collision processes. Simulations of picosecond pulse propagation in optical fibers showed that either third-order dispersion or Raman scattering can generate the rogue soliton independently.

In 2011, S. Vergeles and S. K. Turitsyn [100] presented a theoretical analysis of rare events of high-intensity fluctuations (optical rogue waves) which occur in fiber communication links using bit-overlapping transmission. They pointed out that in optics the random localized high-intensity pulses might be observed even with the standard Gaussian or even rarer-than-Gaussian statistics, without imposing the condition of an increased probability of extreme value events.

In 2011, F. T. Arecchi et al. [101] studied the linear light propagation in a glass fiber and displayed that the optical rogue waves might be originated from granularity and inhomogeneity. While in nonlinear systems, nonlinearity plays the role of the two ingredients of granularity and inhomogeneity.

In 2011, Q. Tian et al. [102] demonstrated theoretically a method to control the dynamics of optical rogue waves in nonlinear optical fiber with group-velocity dispersion, cubic nonlinearity and linear gain managements. With proper coefficients, optical rogue waves can be restrained, vanished or sustained forever.

In 2011, D. Buccoliero et al. [103] investigated the formation of rogue waves in soft glass tellurite fibers numerically. Their results implied that rogue waves will not occur in any supercontinuum which is limited by loss, such as commercial silica fiber-based supercontinuum sources.

In 2011, N. Akhmediev et al. [104] found that the detailed triangle spectrum of rogue wave may be tested at the early stage of formation in a chaotic wave field. Before the occurrence of dangerous value of wave peak, people may find the rogue wave through the continuum optical spectrums in different sections of wave field.

In 2012, A. Antikainen et al. [105] performed a detailed analysis of the phase sensitivity of optical fiber soliton collision and energy exchange dynamics reduced by Raman scattering. Because of collisions between emerging solitons with a zero-phase difference, they concluded that the extreme soliton trajectories can be identified as a class of optical rogue wave.

In 2012, A. Zaviyalov et al. [106] reported the theoretical observation of optical rogue waves in a dissipative system, namely, a mode-locked fiber laser.

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In 2013, C. Larsen et al. [107] showed that the continuous wave pumped supercontinuum generation depended strongly on the zero-dispersion wavelength of the fiber owing to the low peak power.

In 2013, S. Wabnitz [108] described the optical pulse shoaling in the normal dispersion regime of optical fibers. They demonstrated that the exact solutions of the optical nonlinear shallow water equation are relatively in good match with the numerical solutions for the NLS equation.

In 2013, D. B. A. Karsaklian et al. [109] reported the feedback strength on extreme events in an optical system with time-delayed dynamics.

In 2013, M. Bock and R. Grunwald [110] introduced a class of highly localized wave packets as a generalization of optical Bessel-like needle beams.

In 2014, C. Lecaplain and P. Grelu [111] investigated the rogue wave among noise-like pulse laser emission through experiments. Spectral rogue waves were recorded for both positive and negative dispersions, while the pulse-energy rogue waves were exclusively detected in the anomalous dispersion regime [111].

In 2014, W. P. Zhong [112] demonstrated the controllable parabolic-cylinder optical rogue waves in certain inhomogeneous media. An analytical rogue wave solution of the generalized NLS equation with spatially modulated coefficients and an external potential in the form of modulated quadratic potential was obtained by the similarity transformation.

## 1.7 Financial Rogue Waves

In 2010, Z. Y. Yan [34] first performed the financial rogue wave solutions for a nonlinear option pricing model, which is the nonlinear wave alternative of the Black–Scholes model. These rogue wave solutions may be used to describe the possible physical mechanisms for rogue wave phenomenon in financial markets and related fields. MIT journal *Technology Review* carried a special report with the title "Econophysicist Predicts Rogue Financial Waves." In addition, the United States *Popular Science* reported this work with the title " Econophysicist Claims Rogue Waves Could Account for Volatility in Financial Markets."

In 2011, evolution of the vector financial rogue waves for the coupled nonlinear option pricing model was analyzed by Yan [113].

## 1.8 Nonautonomous Rogue Wave Solutions

As mentioned above, the classic NLS Eq. (1.2.2) has the rogue wave solutions. Theses solutions are localized in both time and space, because of the constant coefficients of the nonlinear models. Rogue wave has the great amplitude near the origin, while it will tend to plane wave when time and space tend to infinity.

In fact, models with coefficients varying in time and space can reflect the propagation of nonlinear wave in the self-similarity medium. The natural question is whether

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the rogue wave solutions can exist in nonlinear model with varying coefficients. If there can, how do their physical mechanisms work?

In 2010, Yan [114] first gave the concept of "rogon." The analytical nonautonomous rogons have been reported for the inhomogeneous NLS equation with variable coefficients in terms of rational-like functions. The elastic interaction between rogons and the influence of the varying parameters was also displayed.

In 2010, 3D time-dependent rogue waves in nonstationary parabolic potentials were obtained and analyzed by Yan et al. [115].

In 2012, Yan and Jiang [116] investigated the nonautonomous discrete rogue wave solutions and their interactions for the generalized Ablowitz–Ladik–Hirota equation with variable coefficients. This equation possesses kinds of discrete models, such as the discrete AL equation, discrete mKdV equation, discrete Hirota equation, etc. The nonautonomous discrete rogue wave solutions were obtained when the external potential function is the linear function of $n$ and coefficients vary with $t$.

In 2013, nonautonomous optical rogue wave solutions for a generalized inhomogeneous higher-order NLS equation with modulating coefficients were discussed by Yan and Dai [117]. Different profiles of rogue waves with various kinds of coefficients were analyzed. In addition, the stability of rogue wave solutions was discussed numerically.

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